

Frozen Jet-Reference Validation

A transparent sanity check of PNMF's production Extra Trees and Random Forest models. Three conventional jet references were frozen before errors were calculated, with every connected aircraft identity removed from training.

94

complete-task jet curves

91 / 3

train / test curves

4.659

RF pooled RMSE, dB

64.423%

RF cells within +/-5 dB

Plain-language result

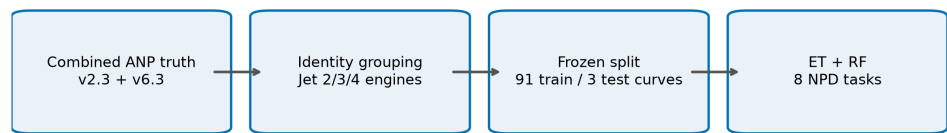
- **RF:** 4.659 dB pooled RMSE and 64.423% of cells within +/-5 dB.
- **ET:** 5.125 dB pooled RMSE and 57.885% of cells within +/-5 dB.
- Category-balanced RMSE is 4.633 dB for RF and 4.840 dB for ET.
- Performance differs materially by category; the four-engine ET reference is the clearest weakness.
- The 1,040 cells are correlated observations inside only **three independent curves**. Threshold agreement is not general accuracy.

Decision boundary

Use this evidence for conceptual screening and model scrutiny. It is not fleet-wide accuracy evidence and not certification evidence.

Data, Selection, and Split

The canonical datastore combines the EASA ANP legacy v2.3 corpus with the v6.3 supplement. The complete-task jet population contains 94 curves in 93 connected aircraft-identity groups.



Noise targets are never used to select references; the physics route remains separate and is not fitted here.

Engines	Engine type	Aircraft identity group	Power row	Distance
2 engines	BR715	Boeing 717-200	717200	0.036467
3 engines	3JT8E5	Boeing 727-200	727EM2	0.023327
4 engines	PW4056	Boeing 747-400	747400	0.352528

Descriptor-only rule: distance = sqrt(sum_j (((x_j - median_j) / IQR_j)^2)), using log10(MTOW), log10(total static thrust), and noise chapter. Zero-IQR terms contribute zero; NPD_ID is the tie-break. Noise targets and model errors never select a reference.

Category	Train	Test	Notes
Jet curves	91	3	Strict identity-group separation
Approach rows / task	270	10	Held-out power rows
Departure rows / task	370	16	Held-out power rows
All test cells	-	1,040	104 power rows x 10 distances

Only singleton identity groups are selectable. The mapping is frozen and the implementation fails if datastore drift changes it.

Model Inputs and Learning Routes

Each requested power row becomes a 12-feature vector. The target is the measured NPD level vector at ten standard slant distances from 200 to 25,000 ft.

Feature	Feature description	Feature description
1-3	Engine type one-hot	Jet / turboprop / piston indicators
4	Engine count	Installed engine count
5-6	log10(MTOW), log10(MLW)	Takeoff and landing weight scales
7	MLW / MTOW	Relative landing-weight descriptor
8-9	log10(per-engine and total thrust)	Per-engine and installed thrust
10	Noise chapter	Certification-era descriptor
11	log10(converted row power in lbf)	Requested operating power
12	Throttle	Normalized operating setting

Frozen production learners

Model	Hyperparameters	Model description
Extra Trees (ET)	500 trees; depth 24; max features 0.5; leaf 1	Highly randomized tree ensemble
Random Forest (RF)	200 bootstrap trees; leaf 2	Bootstrap tree ensemble

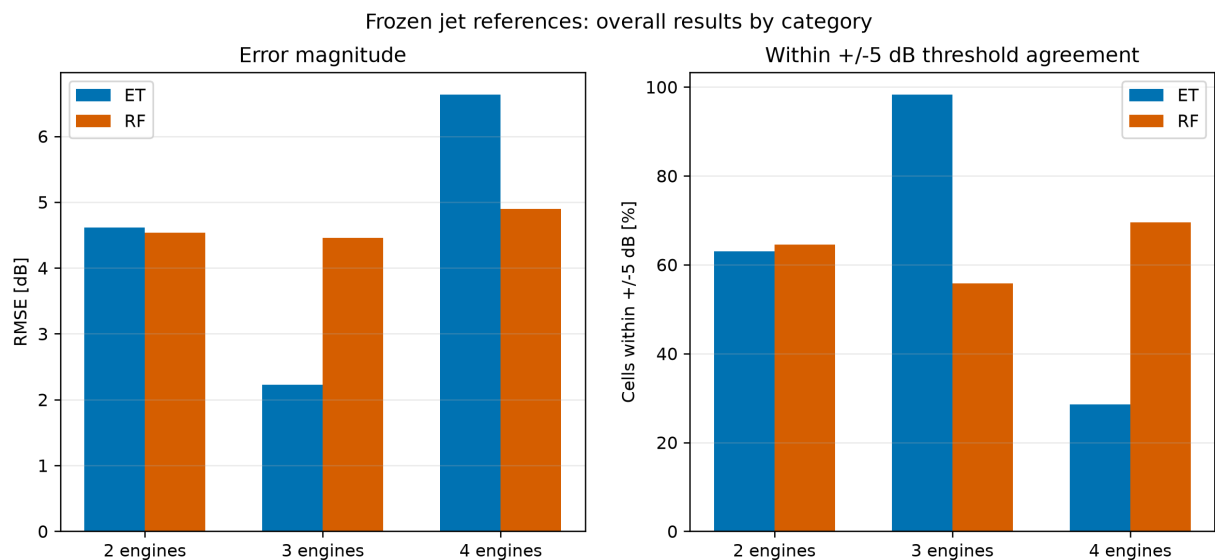
Scoring formulas

```
error = prediction - truth
RMSE = sqrt(mean(error^2))
MAE = mean(abs(error))
within +/-5 = 100 x mean(abs(error) <= 5)
```

Both learned routes use the normal non-increasing distance projection. PhysicsNPDMModel is separate and excluded: it supplies no features or targets to ET/RF.

Overall Measured Results

Model	Aggregation	RMSE	MAE	Bias	RMSE (bal)	Within +/- 5 dB
ET	Cell pooled	5.125	4.499	-1.193	7.366	57.885%
ET	Category balanced	4.840	4.147	-1.265	6.430	63.278%
RF	Cell pooled	4.659	4.304	-0.992	6.427	64.423%
RF	Category balanced	4.633	4.252	-1.375	6.575	63.278%

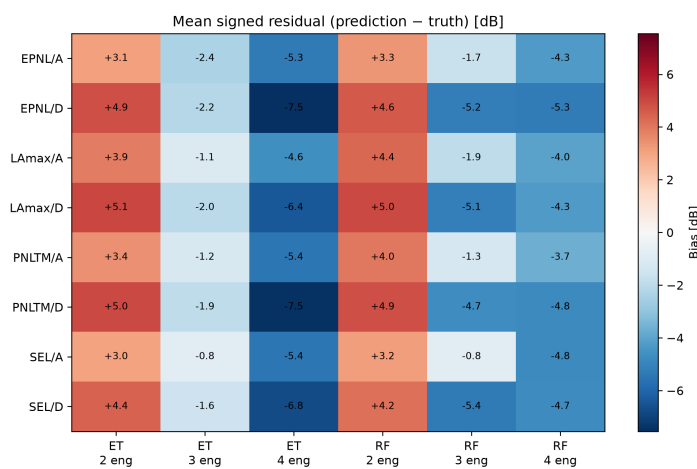


Interpretation: RF has the lower pooled RMSE. Balanced RMSE is closer because every engine-count category contributes equally. RMSE emphasizes large misses; MAE describes typical absolute cell error; bias is prediction minus truth. The threshold percentages describe correlated cells, not independent aircraft successes.

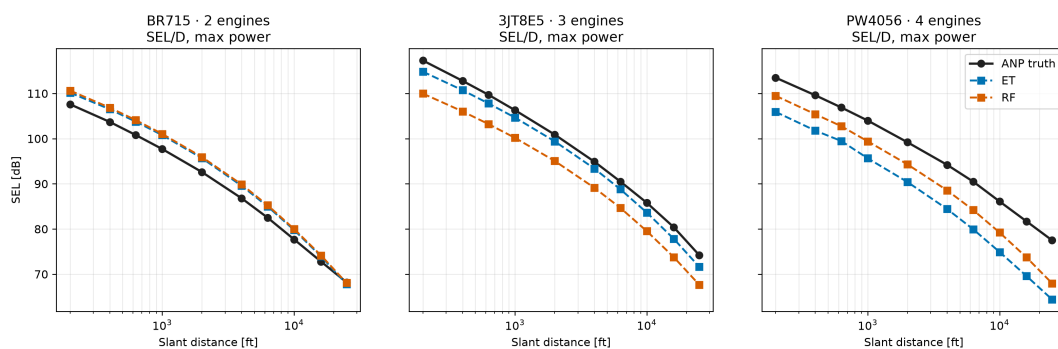
Category and Curve Differences

Category	Engines	ANP	ET	Bias	RMSE	Accuracy
ET	2	4.611	4.269	+4.248	6.610	63.000%
ET	3	2.229	1.863	-1.733	3.228	98.333%
ET	4	6.636	6.310	-6.310	9.452	28.500%
RF	2	4.534	4.294	+4.288	6.255	64.500%
RF	3	4.459	3.914	-3.867	6.334	55.833%
RF	4	4.893	4.547	-4.547	7.136	69.500%

The four-engine reference is the main ET weakness: -6.310 dB mean bias and 28.500% of cells within +/-5 dB. RF also underpredicts it, but less.



Actual versus predicted NPD curves (illustrative task)



The lower chart is an illustrative maximum-power SEL departure slice. All eight tasks, power rows, and ten distances remain in predictions.csv.

Validity Boundaries and Audit Trail

The result is a pre-frozen implementation sanity check on three conventional jet curves with strict identity separation.

Limitations

- Three independent curves are too few for fleet-wide accuracy or certification.
- The 1,040 cells are repeated power-distance observations and are correlated.
- Selection represents only weight, total thrust, and noise chapter.
- The references do not establish performance for unseen families, unusual geometry, or unconventional propulsion.
- Category differences matter; pooled values can hide four-engine weakness.

Physics remains separate

PhysicsNPDMModel is an independent component-source cross-check for SEL and LAmax with explicit bypass-ratio/component assumptions and frozen calibration. It is not trained or evaluated here and does not feed ET/RF.

Reproducible evidence

Seed 20260724. Saved artifacts include selection_candidates.csv, reference_metadata, split.csv, predictions.csv, fit_runs.csv, summary.csv and summary.json, source_manifest.csv, and run_manifest.json under outputs/model_validation/jet_reference.

Final interpretation

RF is stronger on the pooled headline in this frozen check, while balanced results are closer. Treat the category table and residual patterns as the main diagnostic evidence. Use the report for conceptual screening and model scrutiny - never as a certification or broad fleet-performance claim.